

Business Problem Statement

A company wants to better understand **customer behavior and the factors associated with customer churn** in order to improve customer retention and increase long-term customer value. Management has observed differences in churn rates across customer groups and wants to identify the characteristics, behaviors, and service-related conditions associated with a higher likelihood of customers leaving.

The company is particularly interested in factors such as **customer demographics, tenure, usage frequency, support interactions, payment delays, last interaction, subscription type, and contract length**. Identifying the factors and customer segments associated with higher churn risk can help the business prioritize retention efforts and make more data-driven decisions.

You are tasked with analyzing the company's **customer churn dataset** to answer the following overarching business question:

“How can the company leverage customer data to identify factors associated with churn, detect high-risk customer segments, and develop effective strategies to improve customer retention?”

Deliverables

- 1. Data Preparation & Modeling (Python):**
Clean, validate, and transform the customer dataset to ensure data quality and prepare it for analysis.
- 2. Data Analysis (SQL):**
Use PostgreSQL and SQL to analyze customer churn rates, customer behavior, demographic patterns, subscription and contract characteristics, and identify high-risk customer segments.
- 3. Visualization & Insights (Power BI):**
Build an interactive dashboard that visualizes customer churn, churn drivers, high-risk segments, and business impact to support data-driven decision-making.
- 4. Report and Presentation:**
Summarize key findings, interpret their business implications, and provide actionable recommendations to reduce churn and improve customer retention.
- 5. GitHub Repository:**
Include all Python notebooks/scripts, SQL queries, cleaned datasets, and Power BI dashboard files in a well-structured repository.